

Lecture 2

26 May 2026 18:06

Finite Automata $\rightarrow (Q, \Sigma, \delta, q_0, F)$

$\swarrow$ fin. states $\downarrow$ alphabet $\nearrow$ start state $\searrow$ set of accept states

$\delta: Q \times \Sigma \rightarrow Q$

DFA.

$$\delta(q, a) = r$$

Regular Languages $\rightarrow A$ is a reg. lang.

Then $\exists$ a DFA 'M' s.t. M 'recognizes' A

Thm 1] A_1, A_2 are regular languages $\Rightarrow A_1 \cup A_2$ is regular.

$$A_1 \cup A_2 = \{ q \mid q \in A_1 \text{ or } q \in A_2 \}$$

Proof: $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognizes A_1

$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ recognizes A_2

We have to construct $M = (Q, \Sigma, \delta, q, F)$

recognizing $A_1 \cup A_2$.

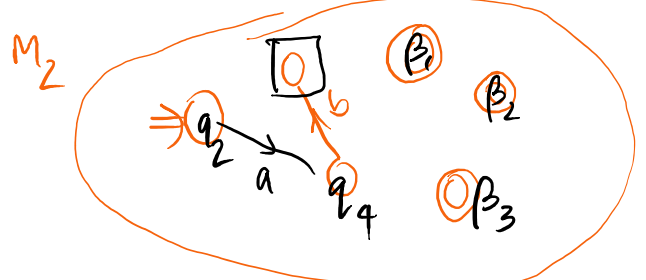
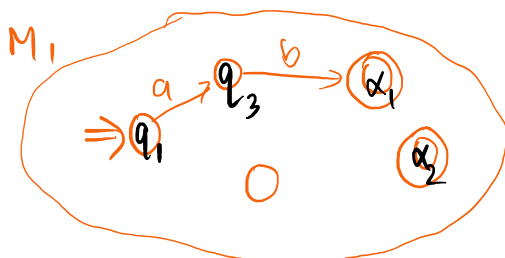
$\rightarrow$ M should accept w if either M_1 accepts w or M_2 accepts w .

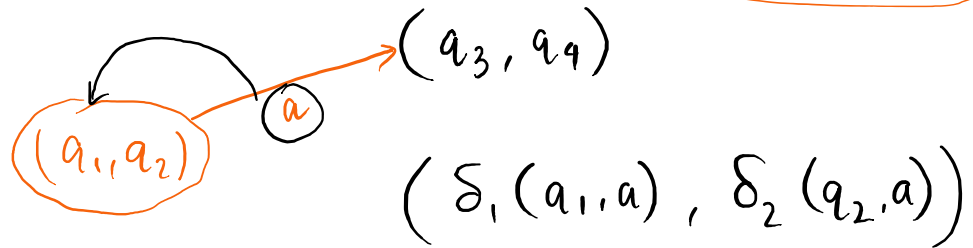
$$Q = Q_1 \times Q_2$$

$$q_0 = (q_1, q_2)$$

$$\delta((q_1, q_2), a) = (\delta_1(q_1, a), \delta_2(q_2, a))$$

$$F = (F_1 \times Q_2) \cup (Q_1 \times F_2)$$





$$\underline{\underline{F_1 \times F_2 = F}}$$

(ab)

$$F = (F_1 \times Q_2) \cup (Q_1 \times F_2)$$

Thm 2) A_1, A_2 reg. languages $\Rightarrow A_1 \circ A_2$ or $A_1 A_2$ is regular.

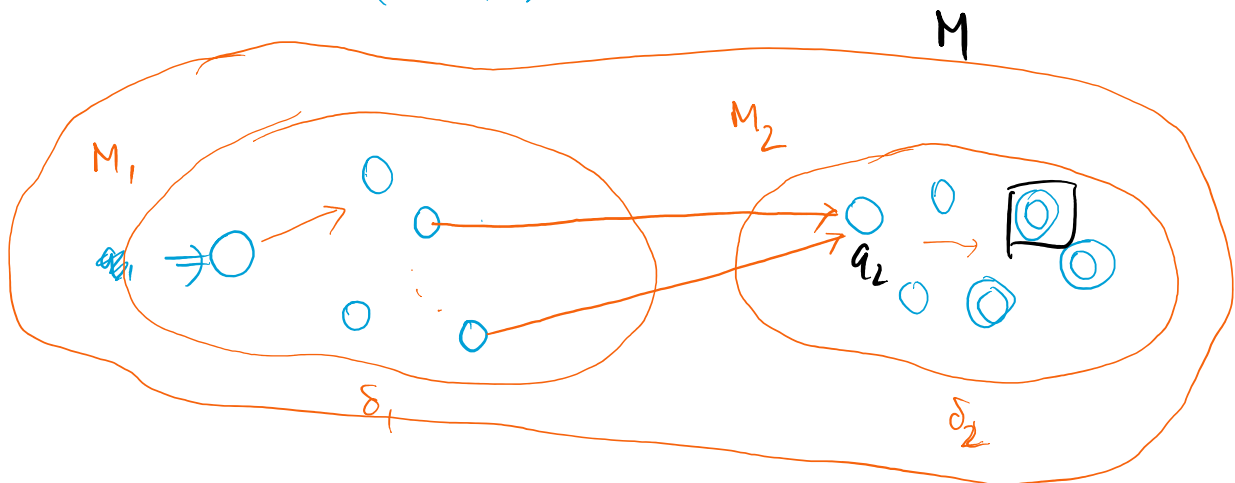
Proof: $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognizes A_1

$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ recognizes A_2

$$A_1 \circ A_2 = \{ w \mid w = xy \text{ where } x \in A_1, y \in A_2 \}$$

$$Q = Q_1 \times P(Q_2)$$

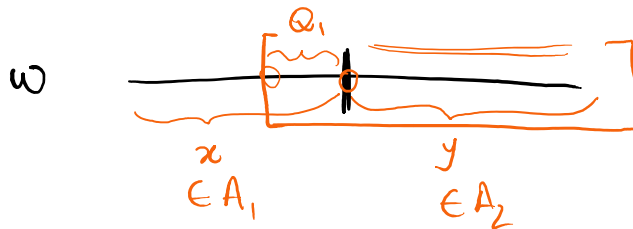
$$q = (q_1, \emptyset)$$



$$Q = Q_1 \cup Q_2$$

$$q = q_1$$

$$\delta(q, a) = \begin{cases} \delta_1(q, a) & \text{if } q \in Q_1 \\ \delta_2(q, a) & \text{if } q \in Q_2 \end{cases}$$



a, aaa

b^*

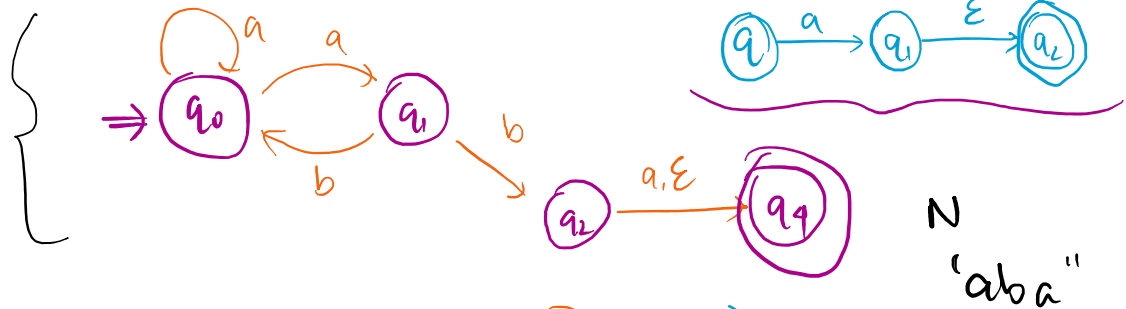
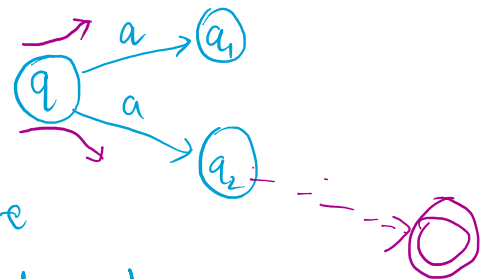
ab

ab

Will be revisited with NFA.

NFA :- Non Deterministic Finite Automaton.

- multiple paths.
- accept if any path leads to an accept state
- ' ϵ '-transition (free move) without reading.



Formally, NFA $\rightarrow (Q, \Sigma, \delta, q_0, F)$

$\delta: Q \times \Sigma_{\epsilon} \rightarrow P(Q)$

$\Sigma \cup \{\epsilon\}$

$\delta(q_0, a) = \{q_0, q_1\}$

$\delta(q_1, a) = \emptyset$

Thm] If an NFA recognizes $A \Rightarrow A$ is regular.

Proof: NFA, $M = (Q, \Sigma_{\epsilon}, \delta, q_0, F)$ recognizes A .

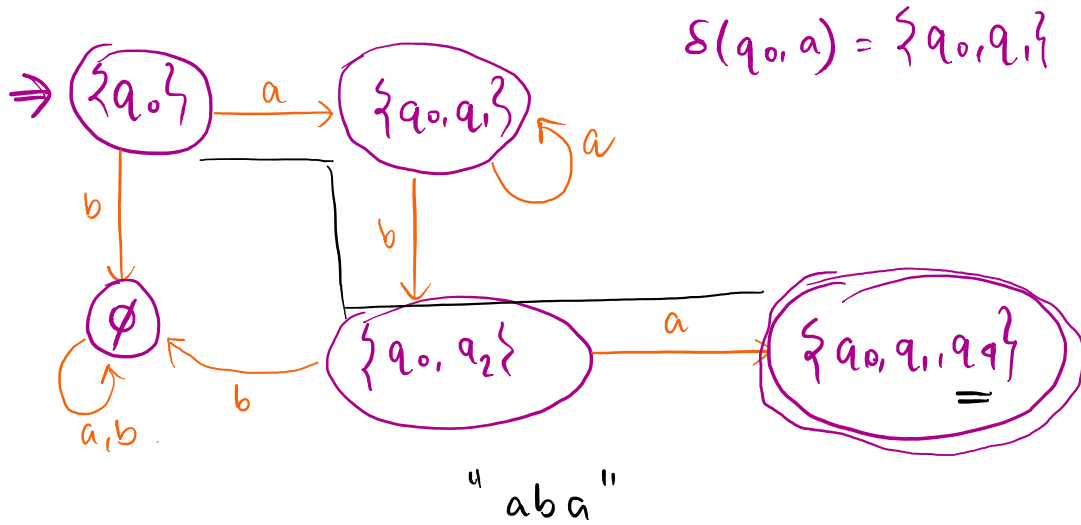
Construct $M' = (Q', \Sigma, \delta', q_0, F')$ recognizing A .

$Q' = P(Q)$

$\delta'(x, a) = \{q \mid q \in \delta(x, a) \text{ for some } x \in x\}$

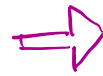
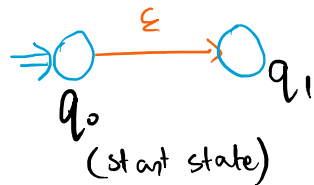
$$\delta'(x, a) = \{q \mid q \in \delta(x, a) \text{ for some } x \in X\} \\ x \in P(Q)$$

$$q_0' = \{q_0\}, \quad F =$$

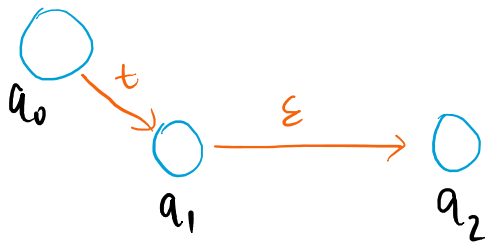


$$F' = \{R \mid R \cap F \neq \emptyset\}$$

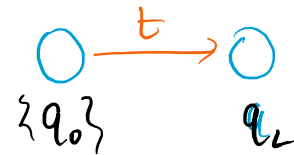
NFA (ϵ -cond.)



DFA (without ϵ)



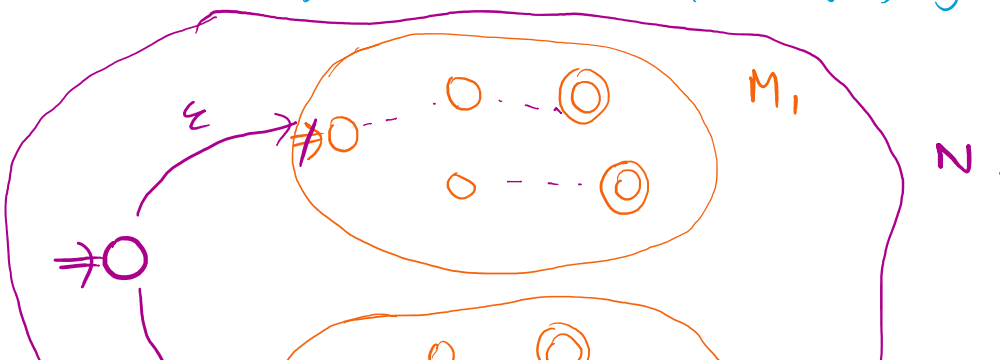
Bypass.

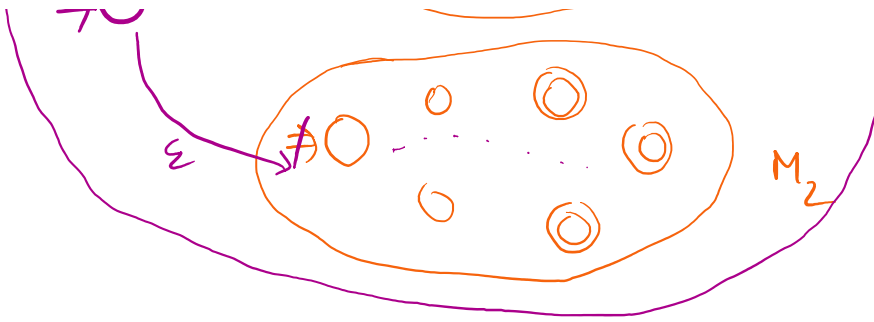


Recall thm] A_1, A_2 reg. languages $\Rightarrow A_1 \cup A_2$ reg. lang.

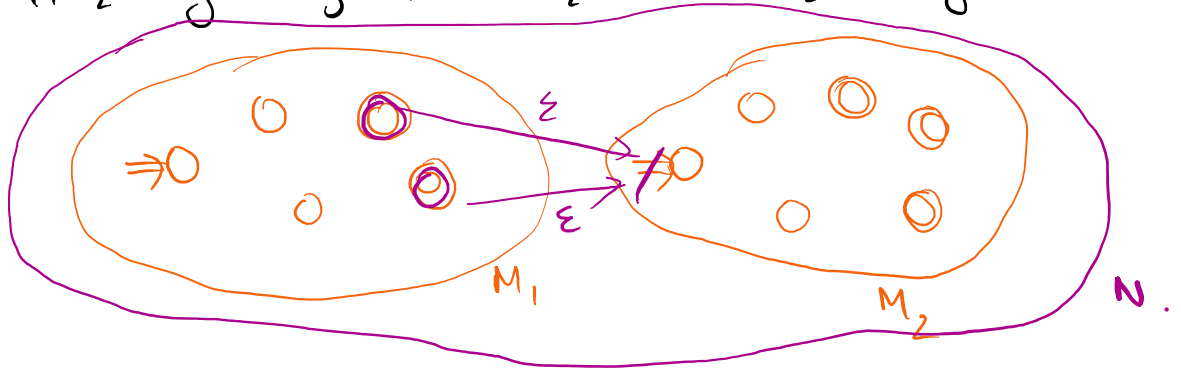
Proof: M_1, M_2 recognizing A_1, A_2 resp.

Construct an NFA. M recognizing $A_1 \cup A_2$





thm] A_1, A_2 reg. lang $\Rightarrow A_1 \circ A_2$ or $A_1 A_2$ is regular.

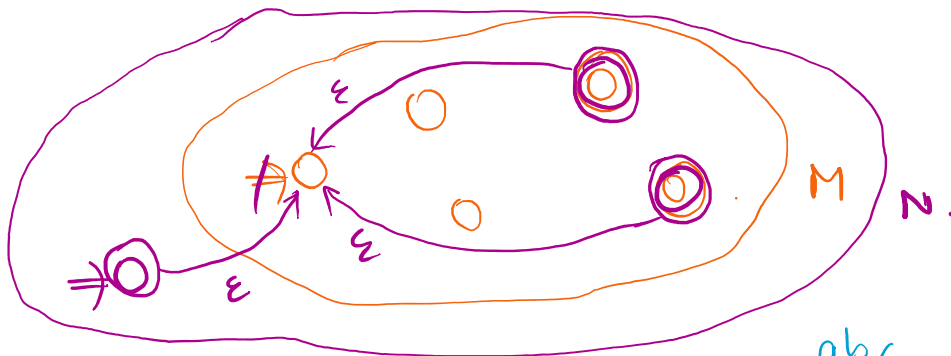


thm] A is reg. language $\Rightarrow A^*$ is also regular.

$$A^* = \{x_1 \dots x_k \mid x_i \in A \ \forall i, k \geq 0\}.$$

$\varepsilon \in A^*$

$\forall A$ language



0,1.

abc.

a

str = ε



N accepts A^* .

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